

Cheng-Nan Liu

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 [Cheng-Nan Liu](#) |  [Cheng-Nan Liu](#)
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OBJECTIVE

A highly motivated fourth-year Ph.D. student with extensive experience in data science. Proficient in time series data analysis and machine learning applications in Seismology. Proven track record of developing advanced machine learning models for hydrothermal systems monitoring and prediction. Skilled in both independent research and collaborative projects, with multiple publications in leading geophysical journals.

INTEREST

Hydrothermal System Monitoring | Ambient Noise Tomography | Machine Learning

EXPERIENCE

- Lawrence Berkeley National Laboratory** May 2024 - Aug 2024
Research Intern, Full time
Berkeley, U.S.A.
 - Develop a machine learning model for image restoration on time series data.
 - Fostered strong collaboration in a multidisciplinary team, enhancing cross-departmental synergy and innovation.
- Institute of Earth Sciences, Academia Sinica** Jan. 2020 - Jul. 2021
Research Assistant, Full time
Taipei, Taiwan
 - Develop a machine learning model and implement automated pipelines for earthquake relocation systems.
 - Develop a physical model using time series data analysis and Bayesian Markov-Chain Monte-Carlo modeling.
 - Worked closely with an international team to integrate and automate processes.

EDUCATION

- University of Utah** Aug. 2021 - Present
Ph.D. candidate, Department of Geology and Geophysics, Advisor: Dr. Fan-Chi Lin
Salt Lake City, UT, U.S.A.
- National Taiwan University** Aug. 2017 - Jun. 2019
Master, Department of Geoscience
Taipei, Taiwan
- National Cheng Kung University** Aug. 2013 - Jun. 2017
Bachelor, Department of Earth Science
Tainan, Taiwan

PROJECTS

- Project A: Wavefield Inpainting Generator** May 2024 - present
Tools: Python, Pytorch, Bash shell, Numpy
 - Led the creation of a Swin Transformer-based algorithm with PyTorch, optimizing over 50 TB of seismic data.
 - Slashed reconstruction time from hours to seconds.
- Project B: Real-time Automated Earthquake Relocation System** Mar. 2021
Tools: Python, Fortran, Bash shell, Numpy
 - Implementing a Unet-based model, doubling the number of detected earthquakes with a 10x increase in speed.
 - Streamlined and integrated multiple technical modules into a fully automated, real-time system.
- Project C: Earthquake Hazard Alarm System** Jun. 2019
Tools: Python, Keras, Bash shell, Numpy
 - Designed a multi-branch CNN in Keras, increasing accuracy by 7% and reducing detection time by 0.3 seconds.
 - Presented findings at conferences, contributing to publications and enhancing the Early Warning System. Work.

PUBLICATIONS AND THESIS

J=JOURNAL, P=IN PREPARATION, T=THESIS

- [J.1] Liu, C.N, et al. (2024). **Multimode ambient noise double-beamforming tomography with a dense linear array: revealing accretionary wedge architecture across Central Taiwan.** *Geophysical Journal International*, 239(1), 467-477.
- [J.2] Liu, C.N, et al. (2023). **Thumping Cycle Variations of Doublet Pool in Yellowstone National Park, USA.** *Geophysical Research Letters*, 50(4), e2022GL101175.
- [J.3] Liu, C.N, et al. (2021). **High-Resolution 3-D Shear Wave Velocity Model of Northern Taiwan via Bayesian Joint Inversion of Rayleigh Wave Ellipticity and Phase Velocity With Formosa Array.** *Journal of Geophysical Research: Solid Earth*, 126(5), e2020JB021610.
- [T.1] Liu, C.N, et al. (2019). **Using low-cost seismometers and machine learning on earthquake early warning.** *Master Thesis*,
- [P.1] Liu, C.N, et al. (in prep.). **Thumping Disturbance of Doublet Pool on May 24th, 2023, in Yellowstone National Park, USA.**
- [P.2] Liu, C.N, et al. (in prep.). **AutoQuake: An automatic system for real-time earthquake relocation.**

SELECTED CONFERENCE PRESENTATION (ORAL)

- **American Geophysical Union (San Francisco, U.S.A. (2023))**
High-Resolution Ambient Noise Tomography across Central Taiwan via Multi-Mode Rayleigh Wave Phase Velocity Inversion
- **Geosciences Annually Assembly (Taoyuan, Taiwan (2023))**
High-Resolution Ambient Noise Tomography across Central Taiwan via Multi-Mode Rayleigh Wave Phase Velocity Inversion.
- **American Geophysical Union (New Orleans, U.S.A (2021))**
High-Resolution 3-D Shear Wave Velocity Model of Northern Taiwan via Bayesian Joint Inversion of Rayleigh Wave Ellipticity and Phase Velocity with Formosa Array.

SELECTED CONFERENCE PRESENTATION (POSTER)

- **American Geophysical Union (San Francisco, U.S.A. (2023))**
Monitoring the Yellowstone Doublet Pool Thumping Cycle through temporal nodal deployments from 2015 to 2023.
- **American Geophysical Union (Chicago, U.S.A. (2022))**
Investigating the Yellowstone Doublet Pool Thumping Cycle through temporal nodal deployments.
- **European Geosciences Union (Vienna, Austria (2019))**
Using Low-cost Seismometers and Machine Learning on Onsite Earthquake Early Warning.

SKILLS

- **Programming Languages:** Python, Fortran, C, C++, Matlab, Bash Shell, C shell
- **Software:** Numpy, Pandas, Pytorch, Keras, Tensorflow, Matplotlib, Opencv, numba
- **Technical:** Time series analysis, Data visualization, Statistical analysis, Inversion
- **Selected courses:** Linear Algebra, Machine Learning, Inversion Theory, Parallel Computing, Physical Design

HONORS AND AWARDS

- **NSF Geothermal INTERN Recipient** 2024
NSF
 - Award funded by NSF's Non-Academic Research Internships for Graduate Students (INTERN) program.
- **The David S. & Inga M. Chapman Fund** 2023
University of Utah
 - The award presented by the University of Utah supports graduate students for professional collaboration.
- **GAGE/SAGE Community Science Workshop Travel Grant** 2023
GAGE/SAGE
 - The award presented by the SAGE/GAGE supports graduate students with professional workshops and hands-on experience.
- **Stokes-Eardley Fellowship** 2021
University of Utah
 - Award given to a student who demonstrates outstanding academic performance and potential in the research.

LEADERSHIP EXPERIENCE & VOLUNTEER EXPERIENCE

- **Seminar Convener** Aug. 2024 - present
University of Utah
 - Organize seminars and invite speakers covering a broad range of topics in seismology/geophysics
 - Fostering academic exchange and collaboration.
- **Conference Worker** May 2021
Academia Sinica, Taiwan
 - Assisting in organizing and running academic conferences.
 - Inviting a diverse range of professionals in Earth Science to promote knowledge exchange and collaboration.
- **Summer School Convener** Jul. 2015 - Aug. 2015
National Cheng Kung University, Taiwan
 - Organize and facilitate summer school programs for high school students.
 - Providing insights into Earth Science and fostering early interest in the field.

TEACHING ASSISTANTSHIPS

- **Geo3100-001: Dynamic Earth** 2023
University of Utah Salt Lake City, UT, U.S.A
- **Geo3030-090: Living with Quakes** 2022
University of Utah Salt Lake City, UT, U.S.A
- **Geo5073: Time Series Analysis** 2019
National Taiwan University Taipei, Taiwan
- **Geo5101: Computer Programming on Geosciences** 2018
National Taiwan University Taipei, Taiwan